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De novo assembly of *Roseomonas mucosa* isolates from healthy human volunteers used to treat atopic dermatitis

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ABSTRACT *Roseomonas mucosa* is a bacterium that is found in the natural microbiota of human skin. Here, we present *de novo* sequence assemblies from *R. mucosa* isolated from the skin microflora of three healthy human volunteers that were used to treat atopic dermatitis patients.

KEYWORDS Gram-positive bacteria, atopic dermatitis, genome

The genus *Roseomonas* was identified in 1993 and included six species (1). *Roseomonas mucosa* is a Gram-negative, coccobacilli-shaped, commensal skin bacterium found in aquatic environments and skin microbiota (2–4). *R. mucosa* can treat atopic dermatitis (AD) (5–7). AD is an inflammatory skin disease causing impaired skin barrier function, immune dysregulation, and susceptibility to *Staphylococcus aureus* infection. *R. mucosa* from healthy human volunteers (HV) not only improved the outcomes of AD models in mouse studies but also reduced the *S. aureus* burden and disease severity in human AD patients (6). We present the genomic sequence of three *R. mucosa* HV isolates used in the Myles et al. study (6). *R. mucosa* was isolated using a phosphate-buffered saline-soaked FloqSwab rubbed on the antecubital fossa and volar forearm. The swab was placed in Reasoner's 2A (R2A) broth containing vancomycin and amphotericin B for 48–72 hours at 32°C. The culture was plated on R2A agar and incubated as before to generate single colonies (7).

R. mucosa, stored at −80°C, was plated on R2A agar and a single colony used to inoculate 100 mL of R2A broth, then incubated at 32°C for 20–25 hours (8). Bacteria were washed and heated to 65°C followed by separate incubations with lysozyme, proteinase K, and SDS. Each sample was incubated with RNaseI at 37°C for 30 min. The mixture was combined with a 3/4 volume of saturated phenol, mixed for 10 min, and centrifugated for 10 min at 12,800 *g*. The aqueous phase was treated with phenol:chloroform:isoamyl alcohol (25:24:1) until the white precipitate interface was absent. A chloroform-only extraction was performed, and the aqueous solution was collected for genomic DNA precipitation. DNA was precipitated with 3 M sodium acetate and subsequently washed 2× with fresh 70% ethanol and resuspended in Qiagen's EB buffer.

PacBio and Illumina sequencing libraries were produced from the same DNA extract using the 10-kb Template Preparation and Sequencing Kit with Covaris g-tubes (insert size of >8,700 bp) and TruSeq Nano DNA Kit using the Covaris M220 (insert size of ~550 bp), respectively (PacBio protocol 100-092-800-06 and TruSeq protocol 15041110 rev. D). Libraries were sequenced using the PacBio Sequel and Illumina MiSeq (2 × 300 bp read length) platforms. Raw Illumina reads were processed with cutadapt (v. 1.12) and quality filtered and trimmed using FASTX-Toolkit (v. 0.0.14). The genomes were assembled using the following programs: CANU (v. 1.0), Bowtie2 (v. 2.2.4), pilon (v. 1.16), and MIRA (v. 4.0). All software was used with default parameters. The assem-

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TABLE 1 *Roseomonas mucosa* genome properties and statistics

Isolate	Raw read BioProject reference	SRA accession no.	Illumina raw reads	PacBio raw reads	PacBio N50	Genbank accession no.	Chromosome size (bp)	No. of chromosomal features	Coverage (mean depth)	%GC	Plasmid sequences	Genbannk accession no.	Plasmid name	Plasmid size (bp)	No. of plasmid features	Coverage (mean depth)	%GC	Total genome features
HV1	PRJNA498096	SRX19883028	3,509,540	1,118,146	4,200	CP034924.1	4,174,791	4,051	1,408	70.5	5	CP034925.1	p1-HV1	387,148	405	761	67.5	
		SRX19883027										CP034926.1	p2-HV1	179,681	221	501	68.0	
												CP034922.1	p3-HV1	51,516	73	559	66.0	4,485
												CP034927.1	p4-HV1	50,949	68	504	67.0	
												CP034923.1	p5-HV1	33,949	39	334	65.0	
HV2	PRJNA386464	SRX19884044	5,688,586	116,074	4,200	CP025061.1	4,231,441	4,087	445	70.5	4	CP025062.1	p1-HV2	424,198	446	384	67.5	
		SRX19884043										CP025063.1	p2-HV2	183,675	214	416	67.5	4,596
		SRX19884042										CP025064.1	p3-HV2	111,031	144	266	68.0	
												CP025065.1	p4-HV2	24,356	43	479	65.5	
HV3	PRJNA386451	SRX19885377	5,925,755	161,064	15,822	CP025060.1	4,244,280	4,109	541	70.7	2	CP025058.1	p1-HV3	440,784	464	480	68.1	4,901
		SRX19885376										CP025059.1	p2-HV3	267,666	328	553	67.3	
		SRX19885375																

bled genomes were custom annotated by Igenbio Inc, while the public annotation was compiled by the NCBI Prokaryotic Genome Annotation Pipeline (9).

The *R. mucosa* genomes consist of a single large circular chromosome (4.17–4.24 Mb) and two to five autonomously replicating plasmid sequences ranging in size from 24.3 to 440.8 kb (Table 1). Genome properties and annotation statistics for each genome are listed in Table 1. Comparative genome analyses will hope to identify genes in *R. mucosa* that aid in its clinical effectiveness against AD.

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DATA AVAILABILITY

The annotated genome of the chromosome and plasmid sequences from the *R. mucosa* HV1, HV2, and HV3 strains has been deposited in Genbank under the accession numbers presented in Table 1. NCBI RefSeq genome assemblies have been generated for HV1 (ASM2622269v1) and HV2 (ASM677102v1). Raw read sequences can be accessed at the NCBI sequence read archive with the following Bioproject numbers (PRJNA498096, PRJNA386464, PRJNA386451). The *R. mucosa* HV1 isolate is available from ATCC.

ETHICS APPROVAL

All human sample collection and processing were performed with the approval of the National Institute of Allergy and Infectious Diseases IRB, which approved the associated clinical trial (NCT02262819). All subjects gave full consent to sample collection and gave written consent to the research protocol.

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